Chapter 2. Permutations and Combinations

In this chapter, we define sets and count the objects in them.

Example

Let S be the set of students in this classroom today. Find $|S|$, the *cardinality* (number of elements) of S.

Section 2.1. Basic Counting Principles

To count $|S|$ we could use ...

The Addition Principle

Dividing students into male students M and female students F, we have

$$|S| = |M| + |F|.$$

The Addition Principle

Definition

A partition of S is a collection of subsets $S_1, \ldots, S_m$ of S such that each element of S is in exactly one:

1. $S = S_1 \cup S_2 \cup \cdots \cup S_m$
2. $S_i \cap S_j = \emptyset$ for $i \neq j$.

The Addition Principle

If $S_1, \ldots, S_m$ is a partition of S, then

$$|S| = |S_1| + |S_2| + \cdots + |S_m|$$

We will see a more sophisticated version of this, called the Inclusion-Exclusion Principle, in Chapter 6.

The Multiplication Principle

If the students in S were sitting in c columns of r rows each, then

$$|S| = r \times c.$$

The Multiplication Principle

If the elements of S are ordered pairs (a, b), where a can be any of x different values, and for each a, b can be any of y different values, then

$$|S| = x \times y.$$

The Subtraction Principle

If R is the set of students registered in the class, and taking attendence, we know the set A of students that are absent, then

$$|S| = |R| - |A|.$$

The Subtraction Principle

Definition

If A is a subset of some universe U, then the *complement of* A is

$$\overline{A} = U \setminus A = \{u \in U \mid u \notin A\}.$$

(Note that the notation $\overline{A}$ is ambiguous, as it doesn't specify U.)

The Subtraction Principle

If A is a subset of some universe U, then

$$|A| = |U| - |\overline{A}|.$$

The Division Principle

Example: If we knew that there were u students in the university, and right now, they were evenly distributed among c classes, then

$$|S| = u/c.$$

The Division Principle

If S is partitioned into m parts of the same size, then the size of any part P is

$$|P| = |S|/m.$$

A more typical example

Example 1: How many two digit numbers consist of two different digits?

[answer]

Section 2.2. Permutations of Sets

Question

How many ways can we order two elements of the set $\{1, 2, 3\}$?

Six ways:

$$12 \qquad 13 \qquad 21 \qquad 23 \qquad 31 \qquad 32$$

Question

How many ways can we order three elements of the set $\{1, 2, 3\}$?

Six ways:

$$123 \qquad 132 \qquad 213 \qquad 231 \qquad 312 \qquad 321$$

Definition

An *r-permutation* of a set is an ordering of r of its elements.
$P(n, r)$ denotes the number of r-permutations of an n element set.

So we saw that

- $P(3, 2) = P(3, 3) = 6$

What is:

- $P(3, 1)$
- $P(n, 1)$
- $P(n, n)$
- $P(n, n-1)$
- $P(n, r)$

[Answers]

Recall 'factorial' notation

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1$$

Theorem

For positive integers $r \leq n$,

$$P(n, r) = \frac{n!}{(n-r)!}.$$

Some Permutation Problems

1. How many three letter words can we make from the letters $\{a, b, c, d, e\}$?
2. How many ways can we arrange 7 men and 3 women in a line so that no two women stand beside each other?
3. How many ways can we arrange 10 people in a line if Jack and Jill cannot stand beside each other.
4. How many ways can we arrange 10 people around a round table?

Circular r-permutations

That last question was asking for the number of *circular* permutations of an n-element set.

Theorem

The number of circular r-permutations of an n element set is

$$\frac{P(n,r)}{r} = \frac{n!}{r \cdot (n-r)!}.$$

1. How many ways can we arrange 10 people around a round table, Jack and Jill cannot be sat together?
2. How many ways can we arrange 5 couples around a round table so that all couples sit together?
3. How many ways can we arrange 5 couples around a round table if all couples are diametrically opposite?

Section 2.3. Combinations of Sets

Combinations are permutations where we don't care about order.

Example

Whereas there were six 2-permutations of $\{1, 2, 3\}$:

$$12 \qquad 13 \qquad 21 \qquad 23 \qquad 31 \qquad 32$$

there are only three 2-combinations:

$$\{1, 2\} \qquad \{1, 3\} \qquad \{2, 3\}$$

- The combination $\{2, 1\}$ is the same as $\{1, 2\}$.
- We usually say *r-subset* instead of *r-combination*.

Notation

The number of r-subsets of an n set is denoted

$$\binom{n}{r}$$

and read "n choose r".

What are $\binom{n}{0}$, $\binom{n}{1}$, $\binom{n}{n}$, $\binom{0}{r}$, $\binom{3}{5}$?

Theorem

For all $0 \leq r \leq n$,

$$\binom{n}{r} = \frac{P(n,r)}{P(r,r)} = \frac{n!}{r!(n-r)!}.$$

[proof]

Corollary

For all $0 \leq r \leq n$,

$$\binom{n}{r} = \binom{n}{n-r}.$$

[proof]

Example

$$\begin{aligned}(x+1)^4 &= (x+1)(x+1)(x+1)(x+1) \\ &= \quad x^4 + \quad x^3 + \quad x^2 + \quad x^1 +\end{aligned}$$

In general

$$(x+y)^n = \sum_{i=0}^{n} \binom{n}{i} x^i y^{n-i}$$

Some permutation questions

1. How many triangles are determined by 12 points in general position in the plane?
2. How many eight-letter words can be constructed using the 26 letters of the alphabet if each word contains three, four or five vowels?
    1. If no letter can be used twice?
    2. If letters can be re-used?

Some more results

Theorem (Pascal's Formula)

For all $1 \leq r \leq n-1$,

$$\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r}.$$

Proof:

$$\begin{aligned}
\binom{n}{r} &= \text{the number } r \text{ subsets of an } n \text{ set} \\
&= \text{the number of } r \text{ subsets not containing element 1} \\
&\quad + \text{the number of } r \text{ subsets containing element 1} \\
&= \binom{n-1}{r} + \binom{n-1}{r-1}
\end{aligned}$$

Theorem

For $n \geq 0$,

$$2^n = \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n}.$$

[proof]

Section 2.4. Permutations of Multisets

Recall the question from Section 2.3:

How many eight-letter words can be constructed using the 26 letters of the alphabet if each word contains three, four or five vowels?

1. If no letter can be used twice?
2. If letters can be re-used?

Here we were choosing letters from the set ALPHABET, and distiguished between choosing elements *with replacement* and *without replacement*. Another way of looking at this is to say that each element occurs in the set multiple times. This doesn't happen in a set.

A *multiset* is like a set, except that elements need not be distinct.

In this section we look at r permutations of multisets.

Notation by Examples

We compactly represent the multiset $\{a, a, a, b, c, c\}$ by

$$\{3 \cdot a, 1 \cdot b, 2 \cdot c\}.$$

If a occurs an infinite number of times in the above set, we write

$$\{\infty \cdot a, 1 \cdot b, 2 \cdot c\}.$$

Question: How many 3-permutations are there of the set

$$\{\infty \cdot a, \infty \cdot b, \infty \cdot c, \infty \cdot d\}?$$

[answer]

Theorem

If S is a multiset containing k distinct elements with infinite repetition, then there are k^r r-permutations of S.

Corollary

If S is a multiset containing k distinct elements each with repetition at least r, then there are k^r r-permutations of S.

Question: How many permutations are there of the following set?

$$\{3 \cdot a, 10 \cdot b, 7 \cdot c, 2 \cdot d\}$$

Theorem

Let S be a multiset containing k distinct elements, the ith of which has repetition a_i. That is,

$$S = \{n_1 \cdot a_1, n_2 \cdot a_2, \ldots, n_k \cdot a_k\}.$$

Then there are

$$\frac{n!}{n_1! n_2! \ldots \; n_k!}$$

permutations of S.

Question: Santa has 10 presents to distribute among three children. Lucy was good, so she gets six of them, and Reid was bad, so only gets one. Hong-cheon gets the other three. How many ways can Santa distribute the presents.

[answer]

Theorem

The number of ways to partition n distinct items into sets of sizes $n_1, n_2, \ldots, n_k$ respectively, where $n = n_1 + \cdots + n_k$ is

$$\frac{n!}{n_1! n_2! \ldots \; n_k!} = \binom{n}{n_1} \cdot \binom{n - n_1}{n_2} \cdot \binom{n - n_1 - n_2}{n_3} \cdot \ldots \cdot \binom{n - n_1 - \ldots n_{k-1}}{n_k}.$$

Section 2.5. Combinations of Multisets

Question: You want to make a fruit basket containing 12 pieces of fruit. You can choose from apples, mangos, plums and those little yellow melons. How many ways can you make up the fruit basket?

Fruits of the same type are indistiguishable!

Theorem

The number of r-submultisets of

$$\infty \cdot a_1, \ldots, \infty \cdot a_k$$

is

$$\binom{r+k-1}{k-1} = \binom{r+k-1}{r}.$$

Question: You want to make a fruit basket containing 12 pieces of fruit. You can choose from apples, mangos, plums and those little yellow melons. How many ways can you make up the fruit basket **which has at least once piece of each type of fruit ?**

Question: What is the number of non-negative integer solutions of the equation:

$$x_1 + x_2 + x_3 + x_4 = 20?$$

- How about $x_1 + x_2 + x_3 + x_4 \leq 20$?
- How about subject to the conditions that $x_1, x_2 \geq 1$ and $x_3 \geq 5$?

Section 2.6. Finite Probability

The counting techniques we have looked at allow us to calculate the odds in many games of chance.

Example

A man in an alley offers the following game. You flip a coin three times. If you get all heads or all tails he gives you three dollars. Otherwise, you give him one.

Should you play the game?

Debate about the wisdom of playing games with a man in the alley, aside...

- There are 8 possible outcomes.
- You win in 2.

So your odds of winning are $1/4$.

- You pay 1 dollar.
- You can win 3. $1 = 3/4$ dollars per one dollar player.

So the payout ratio is $3/1$.

Your expected return on one dollar is $1/4 * 3/1 = 3/4$ dollars, so you shouldn't play.

The Setting

An experiment $\mathcal{E}$ is a random choice of one outcome from a finite sample space S. Each outcome is equally likely.

An event E is a subset of S.

The probability $Prob(E)$ of an event is

$$Prob(E) = \frac{|E|}{|S|}.$$

Example

In an experiment you roll two dice. What is the probability of the event that the dice sum up to 7?
The sample space is set S of possible rolls (a, b) where a is the number on the first die, and b is the number on the second:

$$S = \{(1,1),(1,2),\ldots,(1,6),(2,1),\ldots,(6,6)\}$$

The event that the dice sum to 7 is

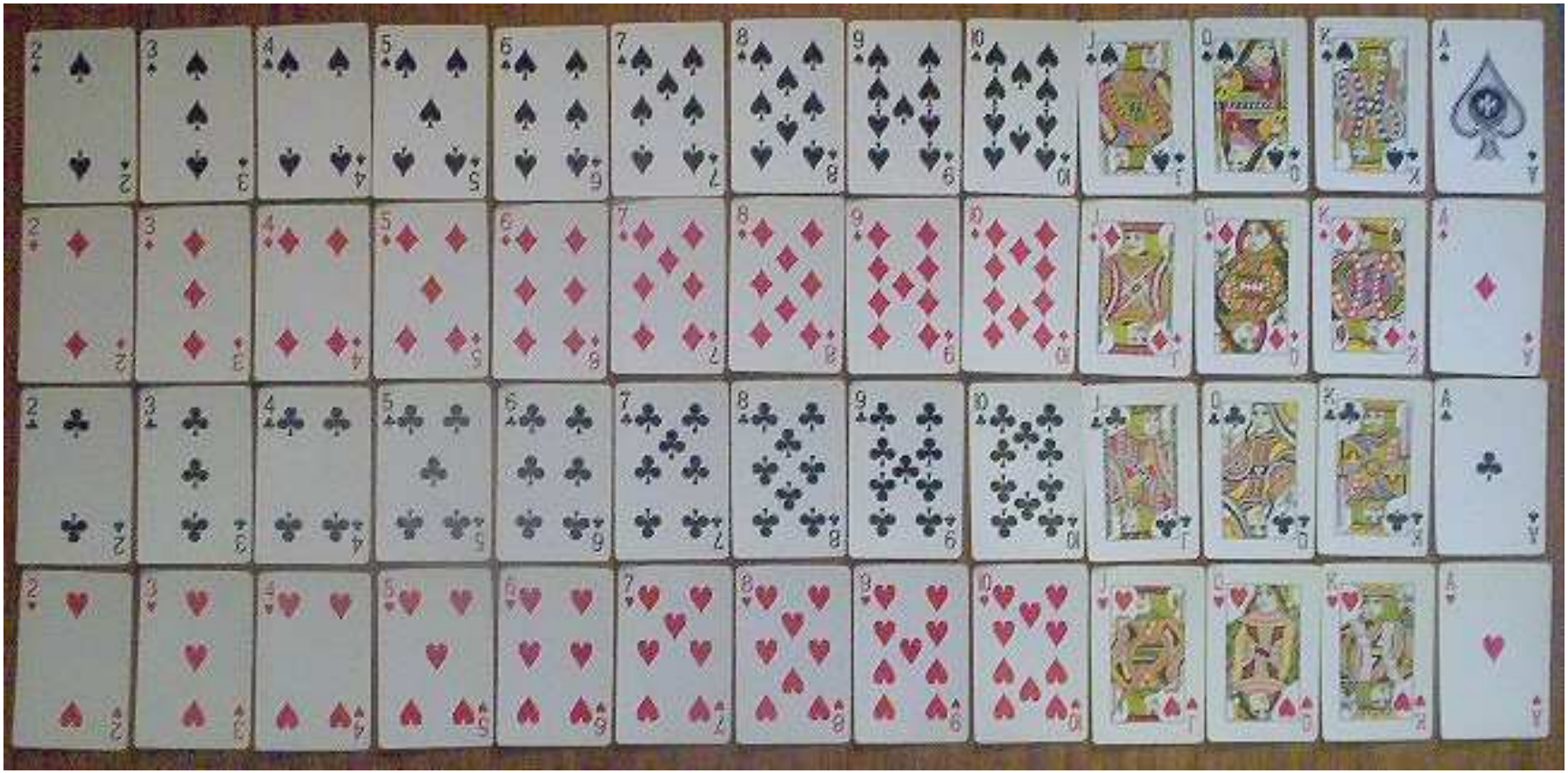
$$E = \{(1,6),(2,5),\ldots(6,1).$$

The probability that the dice add up to 7 is

$$Prob(E) = \frac{|7|}{|36|} \approx .19444$$

Poker

A deck of cards:



Poker

A deck of cards consists of 52 cards.

- Each of four suits: Clubs (C), Hearts (H), Spades (S), Diamonds (D).
- Occur with each ranks 1 (= Ace), 2, 3, . . . , 10, J, Q, K.

Each player is dealt a hand of five cards. The player with the highest hand wins. The hands, in increasing value, are:

1. Pair : two cards having the same rank
2. 2 pairs : two cards of one rank, and two of another
3. 3 of a kind: three cards of the same rank
4. Straight : five cards of consecutive ranks
   (The ace is treated as either 1 or 14.)
5. Flush : five cards of the same suit
6. Full house : three cards of one rank, two of another
7. Four of a kind: four cards of the same rank
8. Straight flush: a straight and a flush

Question: What is the probability of getting a Full House ?

Question: What is the probability of getting none of the above hands?

Question:
You are playing a variation of poker in which you can see three of your opponents cards. He is showing $6, 8$ and 10 of clubs. You have 3 aces. What are the chances you will win?

Section 2.7. Exercises

HW: 2, 6, 10, 21, 39, 47, 63